

Test 1

EEC 3150

02-14-23

For each exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

Test<test number>_Ex<exercise number>_<FirstInitial><LastName>.py

Example: Test01_Ex01_GWest.py

Exercise 1 (25 points)

- a. Write a function `Force(q1,q2,r)` which will return the value of the force between two charged particles with charges q_1, q_2 and separation distance r . This can be calculated using Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

where $k = 9 \cdot 10^9$ (the units are ugly, so I omit them).

- b. Test your function by using it to calculate the force between a proton and electron in a hydrogen atom: $q_{electron} = -1.6 \cdot 10^{-19}C$, $q_{proton} = 1.6 \cdot 10^{-19}C$, $r = 5.3 \cdot 10^{-11}m$.

Exercise 2 (25 points)

- a. Write a function `HMS(n)` which will take an integer n number of seconds and return a string that formats it as a number of hours, minutes, and seconds separated by spaces with units after the values. Here are some examples to illustrate how it should work:

```
>>> HMS(30)
0h 0m 30s
>>> HMS(360)
0h 6m 0s
>>> HMS(3600)
1h 0m 0s
>>> HMS(4000)
1h 6m 40s
```

- b. Demonstrate your function by calling it.

Exercise 3 (25 points)

- a. Write a function `Dot(x,y)` which will return the dot product of two n -dimensional vectors x, y . Reminder:

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$$

- b. Demonstrate your function by creating two lists and calling the function with them as input.

Exercise 4 (25 points)

- a. Write a function `Mode(x)` which will return the mode (most frequently occurring element) of the list. Here is an example:
- ```
>>> x = [1,1,2,2,2,3,3,3,3,3]
>>> Mode(x)
3
```
- b. Demonstrate your function by creating a list calling the function with it.